

Deep Generative Image Modeling: Theory and Applications

Term Project: Facilitate DDPMs in MONAI

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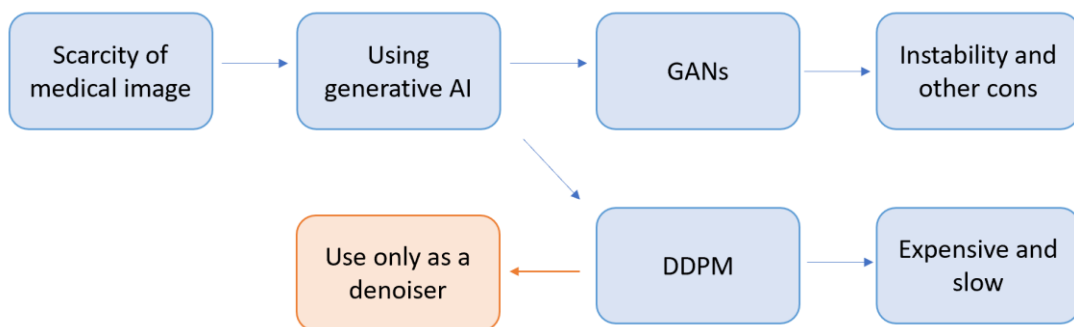
1. Introduction

Generative models are integral to medical imaging, enabling tasks like data augmentation, image reconstruction, and the synthesis of rare conditions. These approaches help overcome data scarcity and variability, ultimately improving diagnostic accuracy.

Generative Adversarial Networks (GANs) have been widely used to produce high-quality medical images. However, GANs often suffer from mode collapse and training instability, making them computationally demanding and challenging to optimize.

Denoising Diffusion Probabilistic Models (DDPMs) represent another class of generative models, each with its own trade-offs. By progressively adding and removing noise, DDPMs can achieve stable and diverse outputs. Yet, their iterative nature requires numerous steps to generate a single image, making them computationally costly and slow.

To mitigate these inefficiencies, we propose using noised images instead of random noise as inputs during image generation, effectively transforming DDPMs into denoisers. This approach circumvents the prolonged training times typically associated with DDPMs while still leveraging their advantages of stable and diverse image generation.



2. Review of Previous Work

Denoising Diffusion Implicit Models (DDIMs) were developed to enhance the efficiency and practicality of Denoising Diffusion Probabilistic Models (DDPMs). DDIMs achieves the reduction in training time through several key modifications:

1. **Deterministic Sampling Process:**

Unlike DDPMs, which use a stochastic process for both forward and reverse diffusion, DDIM employs a deterministic approach during the reverse diffusion phase. This eliminates the randomness inherent in DDPMs, allowing the model to predict the denoised image more directly and efficiently.

2. **Non-Markovian Forward Process:**

DDIM alters the forward diffusion process to be non-Markovian by removing the dependency on the previous timestep. This change enables the model to utilize information from all previous steps simultaneously, facilitating faster convergence during training.

3. **Fewer Diffusion Steps:**

By optimizing the diffusion schedule, DDIM significantly reduces the number of required diffusion steps. This streamlined process means that fewer iterations are needed to achieve high-quality image generation, directly decreasing the overall training time.

4. **Parameter Sharing and Efficient Computation:**

DDIM incorporates parameter sharing across different timesteps and leverages more efficient computational techniques. This reduces the computational overhead per training step, allowing the model to train faster without sacrificing performance.

3.1 Summary of the technical solution

1. Dataset Selection and Preparation:

Utilize the **Medical Open Network for AI (MONAI)** framework to access the **Medical MNIST (MedNIST)** dataset.

2. Model Training:

Train a DDPMs model with the training data.

3. Noise Augmentation:

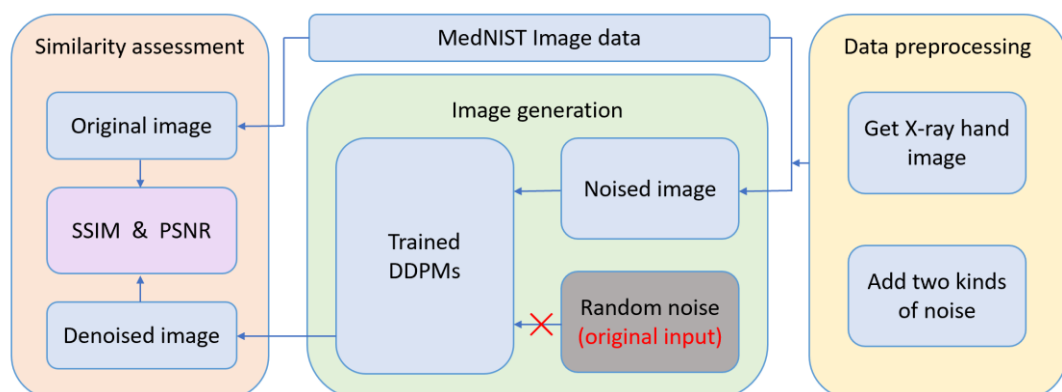
Add noise to the dataset.

4. Image Generation:

During image generation, input the noisy images from step 3 instead of random noise.

5. Evaluation Metrics:

Assess the similarity between the generated images and the original images using **Structural Similarity Index Measure (SSIM)** and **Peak Signal-to-Noise Ratio (PSNR)**.



Sec 3.2: Details of the technical solution

1. Dataset Selection and Preparation:

We only selected hand X-ray images from the MedNIST dataset. There is total 9000 images inside. 8000 of the images were for training and 1000 of them were evaluation data.

2. Model Training:

We referred to the DDPMs model from the MONAI official website and trained a simple DDPMs model for 20 epochs with the 8000 training data.

3. Noise Augmentation:

Add two types of noise—Gaussian and salt-and-pepper—to the 1000 evaluation data for afterward denoising process.

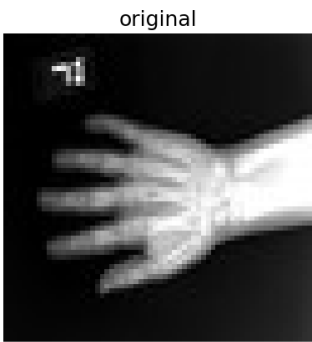
4. Image Generation:

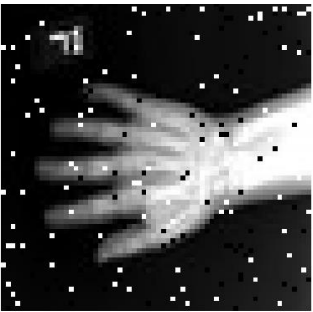
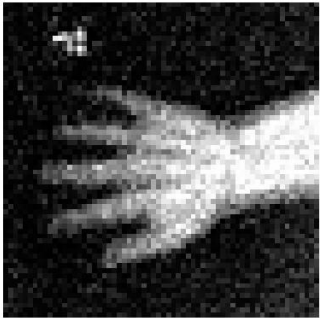
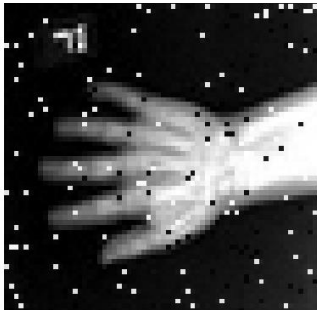
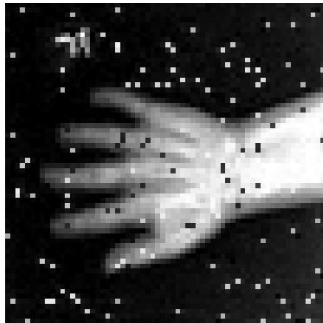
Use the 1000 noised data as the input into the DDPMs instead of random noise. And then let the model iterate through 100 timesteps, generating 1000 denoised image each time.

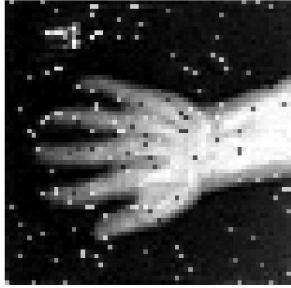
5. Evaluation Metrics:

We operated the similarity assessment by SSIM and PSNR. And we recorded the SSIM and PSNR every timestep to kept track of the similarity during the whole iteration. Lastly, plot out the results and print a sample for each timesteps for direct inspection of the denoise performance.

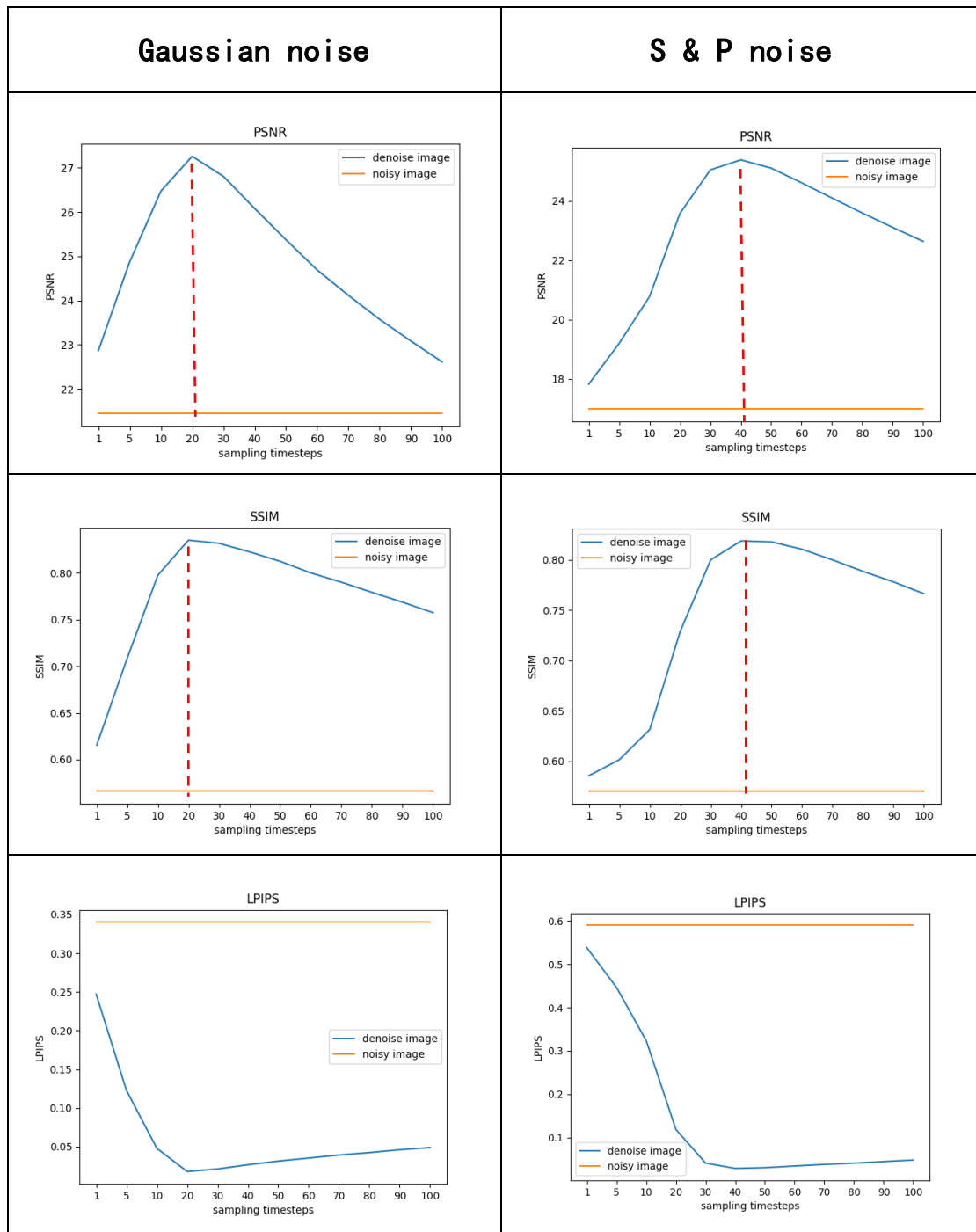
Sec 4: Experiments



	Gaussian noise	S & P noise
noisy		
Timestep 1		
Timestep 5		

Timestep 10		
Timestep 20		
Timestep 40		
Timestep 90		

We printed out the generated images within 100 timesteps. Comparing them with the original image, we can tell that at roughly timestep 20 the gaussian noise was cleared away and at roughly timestep 40 the S & P noise was cleared away.



Then we calculated the PSNR, SSIM and LPIPS at every timestep, and then plotted them out. We can tell that the similarity dropped after a certain peak since the DDPMs would keep adding noise into the image.

Sec 5: Conclusions

In conclusion, this method allows us to generate new types of images without the need to retrain a DDPM from scratch. Instead, we can use any pre-trained DDPM for denoising and image generation. By doing so, we can reduce the number of timesteps from the usual hundreds to under 50, significantly speeding up the process. Since DDPMs are designed for denoising, the results are still quite effective, maintaining high quality in the generated images. This makes the approach both efficient and practical for medical imaging applications.

Sec 6: References

- [Project-MONAI/MONAI: AI Toolkit for Healthcare Imaging](#)
- [mednist_GAN_tutorial.ipynb – Colab](#)
- [autoencoder_mednist.ipynb – Colab](#)
- [Mastering Denoising Diffusion Probabilistic Models](#)
[\(DDPMs\) | by LM Po | Medium](#)
- [GenerativeModels/tutorials/generative/2d_ddpm/2d_ddpm_tutorial.ipynb at main • Project-MONAI/GenerativeModels](#)